Natural logarithm

1 Find the value of each of the following correct to four decimal places:

a $\ln 94$ **b** $\ln 0.042$ **c** $\ln 78^4$ **d** $\ln(18 \times 35)$

2 Consider $x = \ln 31$. Find the value of x, correct to two decimal places.

3 Use the properties of logarithms to express each of the following without any powers or surds:

a $\ln y^2$ **b** $\ln \dfrac{1}{y^2}$ **c** $\ln \sqrt{y}$ **d** $\ln \sqrt[3]{y}$

4 Use the properties of logarithms to rewrite $-2 \ln x$ with a positive power.

5 Rewrite each of the following as the sum and difference of logarithms:

a $\ln\left(\dfrac{15}{8}\right)$ **b** $\ln\left(\dfrac{pq}{r}\right)$ **c** $\ln\left(\dfrac{19}{7}\right)$

6 Rewrite each of the following expressions as a single logarithm:

a $\ln 3 + \ln 5$ **b** $\ln 24 - \ln 4$ **c** $\ln 8 - \ln 32$ **d** $\ln(3x) + \ln(5y)$

7 Which expression is equivalent to $2 \ln(7x)$ for $x > 0$?

A $\ln(14x)$ **B** $\ln 14 + \ln x$ **C** $\ln 49 + \ln x$ **D** $\ln(49x^2)$

8 Simplify:

a $e^{\ln w}$ **b** $\ln(e^x)$ **c** $2^{\ln e^y}$ **d** $e^{\ln x} \times e^{\ln y}$

e $\ln\left(\dfrac{e^x}{e}\right)$ **f** $\ln e^x - \ln e^y$ **g** $\ln\left(\dfrac{1}{e^x}\right)$ **h** $\ln\left(\dfrac{e^5}{e^x}\right)$

9 Evaluate each of the following expressions:

a $\ln e^{3.5}$ **b** $\ln e^4$ **c** $\sqrt{6} \ln\left(e^{\sqrt{6}}\right)$ **d** $\ln\left(\dfrac{1}{e^2}\right)$

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